

UĞURHAN GÜL

Senior Full Stack .NET Developer | Technical Lead | Software Architecture

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PROFESSIONAL SUMMARY

Senior Full Stack .NET Developer and Technical Lead with 12+ years of professional software development experience, specializing in enterprise applications, distributed systems, microservices, event driven architectures, high volume data processing, and software architecture.

Delivered and led nearly 100 enterprise projects across 13 industries, contributing to large scale platforms for organizations including Turkcell, Hepsiburada, Vodafone, Ziraat Bank, TOFAŞ/FIAT, Hyundai, Arçelik, and AssisTT.

Strong hands on expertise in C#, .NET, ASP.NET Core, Web API, Entity Framework Core, PostgreSQL, SQL Server, Kafka, RabbitMQ, MongoDB, Redis, Docker, CI/CD, and distributed system architecture.

Experienced in both hands on software engineering and technical leadership, including architecture design, performance optimization, code review, engineering standards, mentoring, security remediation, and end to end technical delivery.

Recent work also includes integrating LLM and vision language technologies, AI assisted document processing, Model Context Protocol servers, self hosted AI inference, and intelligent automation into production oriented .NET systems.

Career Highlights

- Migrated 150M+ records with zero data loss using a custom .NET migration engine
- Reduced projected enterprise migration time by approximately 70%
- Led engineering teams of up to 13 developers
- Delivered 98 enterprise projects across 13 industries
- Designed architecture supporting 25+ banking websites across 10+ countries
- Developed reusable platforms used across 100+ enterprise client websites
- Remediated 30+ critical security vulnerabilities
- Built event driven and multi tenant systems using .NET, Kafka, RabbitMQ, PostgreSQL, MongoDB, and Redis

TECHNICAL SKILLS

Backend Development: C# • .NET 8/9/10 • .NET Core • .NET Framework • ASP.NET Core • ASP.NET Core Web API • ASP.NET MVC • Entity Framework Core • Entity Framework • Dapper • LINQ • Minimal APIs • RESTful APIs • WCF • SOAP Web Services • SignalR • Go • Python

Software Architecture: Microservices • Distributed Systems • Event Driven Architecture • Clean Architecture • Domain Driven Design (DDD) • CQRS • MediatR • Modular Monolith • Multi Tenancy • SOLID Principles • Design Patterns • Dependency Injection • Architecture Decision Records

Data & Messaging: PostgreSQL • SQL Server • MongoDB • Redis • QuestDB • Apache Kafka • RabbitMQ • MassTransit • Large Scale Data Migration • Query Optimization

DevOps & Observability: Docker • Docker Compose • GitHub Actions • TeamCity • CI/CD • Nginx • Prometheus • Grafana • Loki • Serilog • Git

Cloud: Microsoft Azure (App Service, SQL Database, Blob Storage) • Azure DevOps (basic) • On Premises Deployment • Visual Studio Online / VSTS lineage since 2015

AI & LLM Integration: LLM and Vision Language Model Integration • Self Hosted AI Models • OpenAI API • Google Gemini API • Model Context Protocol (MCP) • AI Assisted Workflows • Document Classification • Deterministic Rule Engines (NRules)

Frontend: React 19 • Next.js • TypeScript • JavaScript • AngularJS • Tailwind CSS • HTML/CSS • Umbraco CMS

Testing & Engineering Practices: xUnit • FluentAssertions • Playwright • axe core • Code Review • SDLC • Performance Optimization • Multithreading • Parallel Programming • Technical Documentation

PROFESSIONAL EXPERIENCE

Independent Software Engineer / Consultant | Self-Employed | Jan 2026 – Present

Kocaeli, Türkiye | Remote

Provide independent software engineering, architecture, and product development services focused on enterprise .NET systems, AI integrated applications, distributed architectures, and high performance backend platforms.

Enterprise Aerospace Compliance Platform

- Designed and developed a 13-project .NET 10 modular monolith organised into bounded contexts
- Engineered a document ingestion and classification pipeline extracting 22 document types with bounding box evidence
- Implemented AI assisted extraction and classification workflows for regulated aerospace documentation
- Integrated self hosted AI inference to support confidential and fully air gapped environments, with no CDN, cloud model or telemetry
- Developed deterministic compliance and validation workflows using NRules across 47 rule classes against AS9100 and EN 10204
- Separated concerns so that models extract and locate while the rules engine alone judges, keeping audit verdicts reproducible and pinning every finding to its source page coordinates
- Implemented supply chain document traceability across purchase orders, material documentation, and shipping records
- Implemented PostgreSQL persistence on EF Core 10 with insert only findings and an append only audit log enforced by database triggers
- Built automated testing infrastructure of 2,600+ xUnit and FluentAssertions tests, alongside 31 architecture decision records
- Developed a React 19 / TypeScript operator interface with Playwright end to end and axe core accessibility suites

DropXML / Pax Mercatus – AI-Powered SaaS

- Architected a multi tenant SaaS ecosystem based on 9 event driven .NET services
- Implemented asynchronous messaging using MassTransit and RabbitMQ
- Designed distributed persistence using PostgreSQL, MongoDB, and Redis

- Developed tenant scoping middleware and multi tenant authorization architecture
- Integrated AI services for XML schema detection, product categorization, and content processing
- Developed marketplace integrations for Trendyol, Hepsiburada, and N11
- Designed configurable cascade pricing and product management workflows
- Developed frontend applications using Next.js, React, and TypeScript
- Created shared NuGet infrastructure implementing the Result pattern for reusable application capabilities
- Implemented Docker based deployment environments with Prometheus, Grafana and Loki observability
- Migrated CI/CD workflows from TeamCity to GitHub Actions
- Produced 1,047 commits across 13 repositories
- Extended the work into Pax Mercatus, an SEO analysis backend with seven analyzers behind a multi provider AI layer, Gemini primary with automatic OpenAI fallback

ML-Driven Algorithmic Trading Research

- Migrated a multi strategy Python research environment toward a high performance .NET 10 architecture
- Integrated TorchSharp and CUDA for GPU accelerated machine learning workloads
- Implemented reinforcement learning research workflows using PPO actor critic and dual head DQN agents
- Developed a 47 feature market state building pipeline across multiple timeframes
- Designed custom reward functions evaluating equity curves and drawdown penalties
- Integrated QuestDB for high performance time series storage and retrieval
- Integrated MetaTrader 5 through PythonNet
- Designed the platform as an experimental research system rather than a live trading product

YouTrack MCP Server – Open Source

- Developed an open source Model Context Protocol (MCP) server
- Integrated the platform with the JetBrains YouTrack REST API
- Exposed issue, project, and user management capabilities to AI assistants
- Implemented Zod validation schemas generated at runtime from per project custom field introspection, with rate limiting and exponential backoff retry
- Designed the solution for AI assisted software development workflows

Senior Full Stack .NET Developer | Kafein Technology Solutions | Sep 2024 – Dec 2025

İstanbul, Türkiye | Hybrid

Worked on large scale enterprise projects for Turkcell, Hepsiburada, Vodafone, and AssisTT, focusing on distributed applications, backend engineering, high volume data processing, multi tenancy, and performance optimization.

Turkcell – Large-Scale Data Migration

- Designed and developed a custom .NET 9 migration engine
- Migrated 150M+ rows of user generated content from legacy Drupal/MySQL infrastructure to PostgreSQL
- Completed migration processes with zero data loss
- Implemented configurable multithreaded processing for large volume workloads
- Used Dapper on the bulk path to bypass Entity Framework Core overhead
- Designed PostgreSQL trigger management mechanisms for high volume inserts
- Implemented data validation, fault handling, logging, and migration monitoring
- Reduced projected production migration duration by approximately 70%

Hepsiburada & Vodafone

- Developed multi tenant and content processing services using Go and .NET
- Worked with Kafka and MongoDB within distributed architectures
- Developed a generic ContentBinderPipeline abstraction for content processing pipelines
- Improved scalability and maintainability of backend services
- Cut new widget onboarding from days to hours
- Supported platforms designed for high volume enterprise traffic

AssisTT – Workforce Planning

- Designed a workforce resource planning module from scratch
- Developed CRUD, approval, forecasting, and planning workflows
- Worked across .NET, Erlang, and AngularJS with Entity Framework Core
- Automated complex operational reporting processes
- Automated KPI and workforce planning calculations
- Eliminated more than 20 hours of manual reporting and planning work per week

Additional Contributions

- Developed REST APIs and enterprise middleware components
- Implemented request interception and SEO slug generation services on Turkcell's .NET Aspire microservices platform
- Contributed to performance optimization and scalable application architecture
- Supported enterprise production environments and integration processes

Senior Software Developer / Team Lead | Clockwork Agency | Oct 2019 – Sep 2024

Kocaeli / İstanbul, Türkiye | Remote

Led technical architecture and software delivery for enterprise applications across banking, automotive, retail, hospitality, and other industries.

Technical Leadership

- Led frontend and backend engineering teams of up to 13 developers
- Managed technical delivery across 98 enterprise projects in 13 industries
- Mentored junior and mid level developers
- Established structured code review practices, with every pull request reviewed before merge
- Defined engineering standards based on SOLID principles and maintainable architecture
- Supported architecture decisions, technical estimations, and project planning
- Established and maintained CI/CD processes
- Coordinated technical delivery across multiple simultaneous enterprise projects

Enterprise Platform Architecture

- Architected and maintained the agency's reusable Umbraco/.NET framework
- Supported adoption across 100+ enterprise client websites
- Managed reusable architecture across 8 boilerplate repositories
- Led five major platform version upgrades from Umbraco v8 through v13
- Standardized common application capabilities and development practices
- Reduced typical new project setup time from approximately 2 weeks to 2 days

TOFAŞ / FIAT Online Sales Platform

- Architected backend API infrastructure for online automotive sales, .NET 6 through .NET 8
- Developed shared .NET infrastructure consumed by three brand APIs
- Supported FIAT, Alfa Romeo, and Jeep
- Integrated Iyzico online payment processes
- Designed a 12 state order lifecycle
- Implemented end to end transaction and audit tracking
- Developed reusable backend services and API components

Ziraat Bank

- Owned technical development and architecture for a shared banking CMS ecosystem on SQL Server
- Supported 25+ banking websites across 10+ countries
- Developed reusable Umbraco/.NET components
- Implemented MediatR based request handling
- Developed Payten payment and banking related integrations
- Implemented Bankkart application validation
- Integrated live fund portfolio retrieval
- Improved maintainability through shared core architecture

Security & Compliance

- Led security remediation activities across banking and government projects
- Remediated 30+ critical vulnerabilities
- Developed reusable Content Security Policy implementations
- Implemented encryption and anonymization components
- Developed server side CAPTCHA mechanisms
- Supported penetration testing remediation
- Contributed to applications operating within ISO 27001 scoped environments

DevOps & Engineering Standards

- Established TeamCity CI/CD pipelines
- Improved deployment standardization
- Introduced systematic pull request and code review processes
- Improved development consistency through reusable engineering standards
- Supported developer mentoring and knowledge sharing

Full Stack .NET Developer | Bank Software Solutions | Dec 2015 – Jul 2019

Kocaeli, Türkiye

- Designed and developed enterprise web applications in C# on the Microsoft .NET Framework stack
- Owned development of a maritime agency automation platform from architecture through delivery
- Architected a 7-project C# / .NET modular monolith organised by business capability, covering operations, proforma accounting and document generation, over a shared Entity Framework code first data layer
- Developed RESTful APIs for external system integrations
- Built and consumed WCF services for system to system integration across the platform
- Implemented SQL Server based data architectures
- Implemented OAuth secured endpoints
- Designed a custom formula engine for proforma accounting
- Developed automated document generation functionality
- Automated the production of 30+ official maritime templates
- Supported customs, port authority, crew, cargo, and maritime operational workflows
- Integrated real time vessel tracking services
- Developed Bosphorus Strait monitoring capabilities
- Implemented real time notifications using SignalR

- Reduced reliance on manual operational status tracking

.NET MVC Developer | Pavlike Web Applications and Solutions | Apr 2015 – Nov 2015

Kocaeli, Türkiye

- Developed and maintained web applications using ASP.NET MVC
- Developed backend and database components
- Participated in requirements analysis with customers, managers, and end users
- Investigated and resolved software compatibility and integration issues
- Supported application debugging and maintenance
- Contributed to production issue resolution

Technologies: C#, ASP.NET MVC, Entity Framework, SQL Server, JavaScript, HTML, CSS

.NET Web Developer | Piyasa Group | Aug 2014 – Oct 2014

Kocaeli, Türkiye

- Developed backend web application components
- Developed and maintained database structures
- Reviewed and debugged existing code
- Resolved application defects and technical issues
- Supported application engineering throughout the software development lifecycle

Technologies: C#, ASP.NET MVC, Entity Framework, SQL Server, JavaScript

Web Developer | LuckyEye | May 2014 – Jun 2014

Kocaeli, Türkiye

- Developed and maintained corporate websites
- Resolved scripting and frontend issues
- Developed interfaces using HTML, CSS, JavaScript, and jQuery
- Supported website maintenance and technical troubleshooting

Junior Web Developer | Revo Bilişim | Dec 2012 – Jun 2013

Kocaeli, Türkiye

- Supported web application operations
- Monitored database and application performance
- Investigated software and database performance issues
- Performed technical maintenance and troubleshooting

Software Development Intern | Türkiye İş Bankası | Sep 2011 – Jun 2012

İstanbul, Türkiye

- Gained professional experience within an enterprise technology environment
- Supported technology and software related operational processes
- Developed foundational knowledge of enterprise systems and software development practices

SELECTED PROJECTS

DropXML – Multi-Tenant AI-Powered E-Commerce SaaS

Role: Software Architect / Full Stack Developer | Jan – May 2026

Multi tenant SaaS platform designed to automate e commerce operations for Turkish marketplace sellers.

- Architected 9 event driven .NET services with MassTransit and RabbitMQ messaging
- Used PostgreSQL, MongoDB, and Redis with multi tenant isolation mechanisms
- Integrated AI powered XML schema detection and product categorization
- Developed integrations for Trendyol, Hepsiburada, and N11
- Developed React and Next.js interfaces, Docker based deployment and CI/CD
- Implemented Prometheus, Grafana, and Loki observability
- Produced 1,047 commits across 13 repositories

Technologies: .NET 10 · C# · Microservices · RabbitMQ · MassTransit · PostgreSQL · MongoDB · Redis · OpenAI API · React · Next.js · Docker

Enterprise Aerospace Compliance Platform

Role: Software Architect / Senior .NET Developer | Apr 2026 – Present

- Developed a 13-project .NET 10 modular monolith in bounded contexts
- Designed AI assisted document classification across 22 document types with bounding box evidence
- Integrated self hosted LLM and vision language inference for air gapped operation
- Developed deterministic NRules validation across 47 rule classes against AS9100 and EN 10204
- Implemented document traceability and an append only audit log enforced by database triggers
- Built 2,600+ xUnit and FluentAssertions tests and 31 architecture decision records

Technologies: .NET 10 · C# · PostgreSQL · EF Core 10 · NRules · Hangfire · LLM · VLM · React 19 · TypeScript · xUnit · Playwright

YouTrack MCP Server

Open Source Project | 2025 – Present

- Developed a Model Context Protocol server for JetBrains YouTrack
- Integrated issue, project, and user management APIs
- Designed the server for AI assisted software engineering workflows

- Implemented runtime validation, retry, and rate limiting mechanisms

Technologies: MCP · REST API · TypeScript · Zod · AI Integration · Docker

UDFtör – UYAP Document Handling Mobile App

Personal Project | 2026

- Built an end to end Flutter mobile application for working with UYAP UDF legal documents
- Delivered it in a framework new to me, using an AI assisted development workflow
- Evaluated mobile signature (CAdES) integration across Turkish operator backed signing providers

Technologies: Flutter · Dart · Android · AI Assisted Development · Document Processing

ML-Driven Algorithmic Trading Research

Personal Research | 2024 – Present

- Developed a .NET based deep reinforcement learning research environment
- Implemented PPO actor critic and dual head DQN research workflows
- Integrated TorchSharp and CUDA for GPU accelerated training and inference
- Developed multi timeframe market feature processing across a 47 feature state
- Used QuestDB for time series storage and MetaTrader 5 through PythonNet

Technologies: .NET 10 · C# · TorchSharp · CUDA · QuestDB · Python · PythonNet · Deep Reinforcement Learning

EDUCATION

Anadolu University — Management Information Systems, undergraduate coursework, 2020 – 2024

Kocaeli University — Associate Degree, Computer Software Technology, 2012 – 2015 · GPA 3.27 / 4.00

Kartepe Industrial Vocational High School — Computer Science, 2008 – 2012

CERTIFICATIONS

Software Architecture: From Developer to Architect — LinkedIn Learning, 2026

Advanced Design Patterns: Design Principles — LinkedIn Learning, 2026

ISO 27001 Information Security Management Systems — Kafein Technology Solutions, 2024

ISO 37001 Anti-Bribery Management Systems — Kafein Technology Solutions, 2024

ISO 22301 Business Continuity Management — Kafein Technology Solutions, 2024

LANGUAGES

Turkish: Native · English: Professional Working Proficiency (C1)

CORE KEYWORDS

C# · .NET · .NET Core · ASP.NET Core · Web API · Microservices · Software Architecture · Technical Leadership · Distributed Systems · Event Driven Architecture · Modular Monolith · Kafka · RabbitMQ · MassTransit · PostgreSQL · SQL Server · MongoDB · Redis · Entity Framework Core · Dapper · REST API · WCF · .NET Framework · Domain Driven Design · CQRS · Clean Architecture · Docker · CI/CD · React · Next.js · TypeScript · AI Integration · LLM · MCP · Microsoft Azure · Flutter · Performance Optimization · Multithreading

REFERENCES

References are available upon request.